

1. The Power of Persistence

File handling is a critical feature for any application. It allows your Python scripts to store data permanently and interact with external resources like text logs, configuration files, and images.

Python's `open()` function is the gateway to the filesystem. It creates a **file object** that acts as a bridge between your code and the storage.

2. The open() Function

The `open()` function requires at least one parameter (the filename) but usually takes two: **filename** and **mode**.

```
f = open("demofile.txt", "r")
```

This command tells Python to locate 'demofile.txt' and prepare it for reading.

3. Understanding Open Modes

There are four primary modes for opening a file:

Mode	Action
"r" (Read)	Default. Opens for reading. Error if missing.
"a" (Append)	Opens for adding data. Creates if missing.
"w" (Write)	Opens for writing. Overwrites existing content.
"x" (Create)	Creates new file. Error if it already exists.

4. Text vs. Binary Mode

You must also decide how Python interprets the data inside the file:

- **"t" (Text):** Default. Used for standard text files (.txt, .html, .csv).
- **"b" (Binary):** Used for non-text files like images (.png) or compiled data.

Example: `open("logo.png", "rb")` opens an image for binary reading.

5. Default Behaviors

Python is designed to be concise. If you don't specify a mode, Python assumes **"rt"** (Read Text).

```
f = open("data.txt")
```

is exactly the same as

```
f = open("data.txt", "rt")
```

6. File Paths & Locations

If the file is in the same folder as your Python script, just use the name. Otherwise, provide the full path:

```
# Windows Path
f = open("C:\\myfiles\\test.txt", "r")

# Linux/Mac Path
f = open("/home/user/test.txt", "r")
```

7. Error Prevention

Opening a file in 'r' mode when the file does not exist will trigger a `FileNotFoundError`.

Best Practice: Always verify the file's existence before opening, or wrap your code in a `try...except` block to handle missing files gracefully.

8. Resource Management

Opening a file consumes system resources. It is vital to close the file once you are finished using the `.close()` method.

```
f = open("data.txt", "r")  
# ... do work ...  
f.close() # Free up the resource
```

9. Practical Exercise

Knowledge Check

Question: What is the function used for opening files in Python?

☐ load()

☐ run()

☒ open()

10. File Handling Summary

- ✓ Use `open ( )` to initiate file access.
- ✓ "r" for reading, "w" for writing.
- ✓ "a" for adding data without overwriting.
- ✓ Always specify "b" for images/media.
- ✓ Remember to `close ( )` your files!

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I/O & File Management Series